

Enrollaa

Real-Time Admission Intelligence Platform

Complete Product Documentation & System Design

Version 1.0 | Confidential | 2025

This document covers: Product Overview, Problem Statement, Target Users, UX Flows, Technical Architecture, Database Design, AI System, Monitoring, Agent System, Marketplace, Monetization, API Design, Codebase Structure, Roadmap, Risk Analysis, Scalability, Go-To-Market Strategy, and Future Features.

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Bndlabs

Section 1: Product Overview

1.1 Vision

Enrollaa envisions a Nigeria where no qualified student misses a university admission opportunity due to lack of information, unreliable sources, or fraudulent agents. We are building the most trusted, real-time admission intelligence system on the African continent, starting with Nigeria.

1.2 Mission

To empower every Nigerian student with accurate, real-time, and personalized admission intelligence replacing uncertainty with clarity, and replacing scams with a safe, verified ecosystem.

1.3 What the Product Does

Enrollaa is a mobile-first (React Native) and web (React) platform that:

- Aggregates real-time admission data from all Nigerian universities, polytechnics, and colleges of education
- Tracks cutoff marks, departmental requirements, supplementary lists, and admission statuses automatically
- Delivers personalized alerts and AI-powered admission predictions based on each student's JAMB score, O'Level results, and course preferences
- Provides an internal, company-controlled agent support system with zero tolerance for external agents or out-of-platform payments
- Offers a verified marketplace for purchasing JAMB ePINs, WAEC result PINs, and other educational services
- Monetizes through a transparent subscription model and minimal, non-intrusive ads for free-tier users only

1.4 Core Platform Pillars

Pillar	Description
Trust & Safety	100% internal agents, no external payments, scam-free ecosystem
Real-Time Intelligence	Automated monitoring of 200+ Nigerian university portals
AI-Powered Personalization	Admission predictions and recommendations per student profile
Verified Marketplace	ePIN/PIN purchases directly in-app, no third-party redirects
Accessibility	Mobile-first, works on low-end devices, designed for Nigerian connectivity

Section 2: Problem Statement

2.1 The Nigerian Admission Crisis

Every year, approximately 1.8 million candidates register for JAMB (UTME) in Nigeria. Of these, fewer than 600,000 gain admission to tertiary institutions in the same cycle. The gap between candidates and admitted students is not only a capacity problem it is a massive information problem that costs families time, money, and opportunity.

2.2 Key Problems Identified

Problem A: Missed Deadlines & Scattered Information

Nigerian university admission processes are fragmented across hundreds of institutional websites. Each school publishes cutoff marks, supplementary lists, and post-UTME schedules at different times, in different formats, with zero centralization. Students must manually check dozens of websites, often missing critical windows by hours.

- JAMB releases admission lists at: <https://www.jamb.gov.ng> but school-specific lists are on each school's portal
- Post-UTME screening dates are sometimes published only 5–7 days in advance
- Supplementary admission lists are often published quietly, benefiting only those actively monitoring
- Students in rural areas with limited internet access are disproportionately affected

Problem B: Rampant Fraud from Fake Agents

A shadow industry of fake admission agents has emerged across Nigeria, exploiting the information gap. These agents:

- Charge students ~~₦20,000–~~₦150,000 with promises of guaranteed admission
- Collect payments via personal bank transfers with no accountability
- Provide false information about cutoff marks and admission chances
- Disappear after collecting money, leaving students with no recourse
- Operate primarily on WhatsApp, Telegram, and Facebook impossible to regulate

SCAM IMPACT ESTIMATE: Conservative estimates suggest Nigerian students lose ~~₦2–~~5 billion annually to fake admission agents and fraudulent education service providers.

Problem C: Lack of Personalized Admission Intelligence

Even when students find information, it is generic. There is no platform that tells a student: 'Based on your JAMB score of 210 and your O'Level grades in Science subjects, here is your probability of admission to Computer Science at UNILAG vs. OAU vs. FUTA and here is what you should do right now.'

Problem D: No Safe Digital Marketplace for Education Services

Purchasing JAMB ePINs and WAEC result checking PINs is an area rife with fraud. Students often buy invalid PINs from unauthorized vendors, losing both time and money during critical admission periods.

2.3 Why Existing Solutions Fall Short

Gap	Explanation
No real-time admission tracking	Websites are static, updated manually and infrequently
No personalization	Information is generic not tied to a student's specific profile
No trusted marketplace	No safe in-platform purchase of ePINs or result PINs
No agent accountability	Informal agents operate with zero oversight
No AI prediction	Students have no data-driven way to assess their chances

Section 3: Target Users

3.1 Primary User Segment: JAMB Candidates (Age 15–25)

This is the core user group students who have sat or are preparing for JAMB UTME and are navigating the university admission process.

- Estimated market size: 1.8 million new candidates per year
- Device usage: 80%+ access the internet via smartphone (Android dominant)
- Connectivity: Frequently on 3G/4G; UI must be lightweight and offline-tolerant
- Pain points: Anxiety about admission, vulnerability to scams, information overload
- Motivation: Secure admission to preferred school and course as quickly as possible

3.2 Secondary Segment: Parents & Guardians

Parents actively involved in their children's admission process represent a significant secondary user group and are often the purchasing decision-maker for subscriptions.

- Age range: 35–60
- Tech literacy: Moderate; prefer simple, clear UI
- Primary need: Stay informed, ensure their child is not scammed, track admission status
- Willingness to pay: Higher than students will subscribe for peace of mind

3.3 Tertiary Segment: Repeat/Transfer Students

Students who were not admitted in a previous cycle, or who are seeking transfers, form a year-round engaged audience needing persistent access to admission updates.

3.4 User Personas

Persona	Description
Chidi, 18 (Lagos)	First-time JAMB candidate, 245 score, wants Engineering at UNILAG. Uses Android daily, concerned about missing supplementary list.
Mrs. Okonkwo, 47 (Enugu)	Mother of two candidates. Wants to track both children's admission simultaneously. Will pay for subscription.
Amaka, 21 (Owerri)	Second-time candidate after missed deadline last year. Needs real-time alerts above all else.
Emeka, 19 (Kaduna)	Rural student with inconsistent connectivity. Needs offline-capable dashboard and SMS fallback.

Section 4: Value Proposition

4.1 Core Value Delivered

Enrollaa delivers three fundamental values that no existing platform in Nigeria currently provides simultaneously:

Value 1 Real-Time Certainty

Students receive admission updates within minutes of official publications not days later. Automated scraping monitors school portals 24/7, so users never miss a supplementary list or a post-UTME announcement.

Value 2 Scam-Free Safety

Every interaction happens within a closed, company-controlled ecosystem. There are no external agents. All payments go through the platform (Paystack/Flutterwave). Users can never be defrauded by a fake agent because no external agents exist within the system.

Value 3 Personalized AI Intelligence

The platform does not just show data it interprets data for each student. The AI engine tells a student their exact admission probability per school, recommends alternative courses and institutions, and suggests actionable steps based on current admission trends.

4.2 Value Proposition Canvas

Student Pain Point	Enrollaa Gain Creator
Pain: Missing deadlines	Gain: Real-time push notifications and dashboard tracking
Pain: Scam agents	Gain: Zero external agents; all support is internal and verified
Pain: Generic information	Gain: Personalized admission predictions and recommendations
Pain: Unsafe ePIN purchase	Gain: Verified in-platform marketplace with receipt and delivery
Pain: Anxiety & uncertainty	Gain: Clear admission probability scores and next-step guidance

Section 5: Competitive Analysis

5.1 Market Landscape

The Nigerian education information market has several established players, but none have built a real-time, AI-powered, scam-proof platform. Here is a direct comparison:

Feature	MySchool Nigeria	Allschool NG	Enrollaa
Real-time admission tracking	No manual updates	No blog-style posts	YES automated 24/7
AI admission prediction	No	No	YES per student profile
Verified internal agents	No open forum	No	YES 2 company agents, MVP
In-app ePIN marketplace	No	No	YES Paystack integrated
Personalized dashboard	Partial	No	YES fully personalized
Mobile app (native)	Basic web only	No	YES React Native
Scam protection model	None	None	YES closed ecosystem
Subscription monetization	No	No	YES ₦4,000– ₦5,000/yr

5.2 Competitive Moat

Enrollaa's long-term competitive advantage is built on three defensible moats:

- **Data Moat:** A continuously growing, proprietary dataset of Nigerian admission patterns, cutoff trends, and school-specific requirements that becomes more valuable over time
- **Trust Moat:** A closed, verified ecosystem where zero scams can occur an increasingly valuable promise in a market burned by fraud
- **AI Moat:** Machine learning models trained on Nigerian-specific admission data that improve with every prediction cycle

Section 6: MVP Feature Breakdown

6.1 Feature 1 Course-Based School Search

Users enter their desired course (e.g., 'Medicine', 'Computer Science', 'Law') and the platform returns a ranked list of accredited Nigerian institutions offering that course. Results include:

- JAMB cutoff mark per school per course (current and previous 3 years)
- Post-UTME screening requirements (O'Level grades, subject combinations)
- School type (Federal, State, Private), location, and accreditation status
- Historical admission statistics (how many were admitted vs. applied)
- Link to the official school portal and JAMB CAPS profile

6.2 Feature 2 Personalized User Profile

Upon registration, users complete a profile that drives the entire personalized experience:

- Full name, state of origin, and state of residence
- JAMB score and registration number
- O'Level subject results (WAEC/NECO) with specific grades
- First and second choice courses and institutions
- Preferred geographical location for school
- Notification preferences (push, email, SMS)

6.3 Feature 3 Real-Time Admission Status Tracking

The platform's central feature. After profile setup, each user sees a personal tracking dashboard showing:

- JAMB CAPS status for their registration number
- Whether their school of choice has published its admission list
- Whether their name appears on any published list (when data is available)
- Countdown timers for important deadlines (post-UTME registration, screening dates)
- Status badges: 'Monitoring', 'List Published', 'Check Now', 'Admitted', 'Not Yet Listed'

6.4 Feature 4 Automated Monitoring System

A backend system that continuously scrapes and monitors official Nigerian university websites and JAMB's portal:

- Puppeteer-based scrapers for JavaScript-rendered school portals

- Cron jobs running every 15–30 minutes for high-priority schools during peak admission season
- Change detection algorithm that identifies new admission lists, updated cutoff marks, or new announcements
- Automatic database updates when changes are detected
- Fallback manual review by internal agents for sites that block scraping

6.5 Feature 5 AI Admission Prediction

Using historical admission data combined with a student's profile:

- Admission probability score (0–100%) per school-course combination
- Comparative ranking: 'You have a 78% chance at UNILAG CS vs. 91% at OAU CS'
- Confidence interval based on data availability
- Alternative recommendations when chances are low
- Trend analysis: 'UNILAG CS cutoff has risen 5 points in the last 3 years'

6.6 Feature 6 Smart Dashboard

A personalized home screen showing:

- Admission status cards for first and second choice schools
- AI prediction score with visual gauge
- Today's important deadlines across Nigerian institutions
- Recent admission announcements nationally
- Quick-access buttons: Check My Status, Contact Agent, Buy ePIN

6.7 Feature 7 Notification System

- Push notifications (FCM/APNs) for mobile app users
- In-app notification bell with unread count
- Email notifications for critical updates
- SMS fallback for users with poor internet connectivity (Termii/Africa's Talking integration)
- Notification categories: Admission List Published, Deadline Reminder, Status Change, System Alert

6.8 Feature 8 Internal Agent Support System

A company-controlled support ecosystem (detailed in Section 13):

- Maximum 2 agents in MVP (company-employed, trained, monitored)
- Chat interface within the app no WhatsApp, no external contact

- Response SLA: 2 hours during business hours
- All conversations logged and monitored by admin
- No agent can request or receive payments outside the platform

6.9 Feature 9 Educational Marketplace

- JAMB ePIN purchase: Official ePIN for JAMB profile creation or reprinting
- WAEC result PIN: Official result checking PINs
- Payment via Paystack or Flutterwave only
- Digital delivery: PIN displayed in-app and sent to registered email within 60 seconds
- Transaction history and receipt generation

6.10 Feature 10 Monetization Layer

- Free Tier: Core search, basic tracking, limited predictions, Google AdMob ads
- Premium Tier (~~N4,000~~—N5,000/year): Real-time alerts, full AI predictions, ad-free, agent priority support, marketplace access
- Payment: Paystack subscription billing on web

Section 7: UX Flow Detailed Screen Descriptions

7.1 Landing Page (Web)

The web landing page serves as the entry point for new users and drives subscription conversions.

Top Navigation Bar

- Left: Enrollaa logo (icon + wordmark in primary blue)
- Center: Navigation links Home | Features | How It Works | Pricing | Blog
- Right: 'Login' (ghost button) + 'Get Started Free' (filled green CTA button)

Hero Section

- Headline: 'Never Miss Your Admission Again.' (large, bold, 48–56px)
- Subheadline: 'Real-time admission tracking, AI predictions, and zero scams built for Nigerian students.'
- Two CTAs: 'Start for Free' (green, primary) | 'Download the App' (outlined, secondary)
- Hero illustration: Nigerian student on phone with admission notification overlay
- Social proof strip: '12,000+ students tracked' | '200+ schools monitored' | '99.8% uptime'

Features Section (3-column cards)

- Card 1: Real-Time Tracking icon 'Know the moment your school publishes admission lists'
- Card 2: AI Prediction icon 'See your exact admission probability before you apply'
- Card 3: Safe Marketplace icon 'Buy JAMB ePIN and WAEC PIN safely, in-app'

Pricing Section

- Free tier: Listed features with 'Get Started Free' button
- Premium tier card (highlighted with accent color): Full feature list, ₦4,000/year CTA
- FAQ accordion below pricing

Footer

- Company info | Contact | Privacy Policy | Terms of Service | Social media links

7.2 Onboarding Flow (Mobile App 5 Screens)

Screen 1: Welcome Splash

- Full-screen gradient (primary blue to dark blue)
- Enrollaa logo centered

- Tagline: 'Your Admission, Tracked in Real-Time'
- 'Get Started' and 'I Already Have an Account' buttons

Screen 2: Phone/Email Registration

- Input: Phone number OR email (toggle switch)
- OTP verification via SMS/email
- Privacy notice: 'We never share your data. Read our Privacy Policy.'

Screen 3: Build Your Profile (Step 1 of 2)

- JAMB Score input (numeric keyboard)
- JAMB Registration Number input
- State of Origin dropdown
- State of Preference dropdown (where they want to study)
- Progress indicator at top: [●●○○]

Screen 4: Build Your Profile (Step 2 of 2)

- Course search field with autocomplete (e.g., type 'Med' → 'Medicine and Surgery', 'Medical Laboratory Science')
- First choice institution search with autocomplete
- Second choice institution search (optional)
- O'Level result grades (subject dropdowns with grade selectors)

Screen 5: Notification Permissions + Summary

- Allow push notifications (system prompt + explanation: 'Get instant admission alerts')
- Summary card: Profile snapshot JAMB score, course, schools selected
- 'Take Me to My Dashboard' CTA

7.3 Dashboard Screen

Top Bar

- 'Hello, Chidi' greeting with notification bell (badge count)
- Premium badge if subscribed; 'Upgrade' pill if free

Admission Status Cards (Primary Focus)

- Card 1 First Choice: School name, course, status badge (monitoring/published/admitted), last checked timestamp
- Card 2 Second Choice: Same structure
- 'Add More Schools to Track' + button (premium only)

AI Prediction Widget

- Circular gauge showing probability % (e.g., '74% Chance of Admission at UNILAG CS')
- 'View Full Prediction' link → opens AI Prediction Screen

Today's Deadlines

- Horizontal scrolling cards showing national deadlines for the day
- Each card: School name, deadline type, time remaining

Recent National Updates

- Feed of latest admission events across Nigerian schools (like a news ticker)
- Tap any item for details

7.4 Search Results Screen

Search Bar (Top, Sticky)

- Course name input with live autocomplete
- Filter chips below: 'Federal', 'State', 'Private' | 'All Locations' (dropdown) | 'Sort: By Cutoff'

Results List

- Each result card: School logo (or initials), school name, course name, cutoff mark badge, location tag, 'Track This School' button
- Sponsored result clearly labeled 'SPONSORED' in small text (for ad-supported tier)

Empty State

- Friendly illustration + 'No results for [course]. Try searching a related term.' message

7.5 School Detail Screen

- School banner image (or placeholder gradient)
- School name, type (Federal/State), location, JAMB code
- Tabs: Overview | Courses | Admission History | Reviews
- Overview Tab: Contact info, website link, JAMB accreditation status
- Courses Tab: Full table Course name, cutoff mark, subject requirements
- Admission History Tab: Chart showing cutoff trends over 3–5 years
- 'Track This School' CTA (sticky bottom button)

7.6 Notifications Screen

- List of notifications in reverse chronological order

- Each item: Icon (admission/deadline/system), title, timestamp, read/unread state
- Swipe left to dismiss; tap to open detail
- Filter tabs: All | Admission | Deadlines | System

7.7 Agent Support Interface

User Side

- Chat list screen showing previous conversations
- 'New Conversation' button → topic selector (Admission Help / ePIN Issue / Account Issue / Other)
- Chat screen: Standard chat UI with message bubbles, timestamp, agent name badge
- Warning banner at top: 'Our agents will NEVER ask for payment outside this app. Report any such request immediately.'

Agent Dashboard (Web, Internal)

- Inbox view: Open, In Progress, Resolved queues
- Each conversation: User name, JAMB score, selected schools, issue type, last message
- Chat panel with user info sidebar
- Admin monitoring panel: All conversations visible to admin in real-time

7.8 Marketplace Screen

- Category tabs: JAMB ePIN | WAEC PIN | NECO PIN | Other Services
- Product card: Service name, official logo, price (₦), 'Buy Now' button
- Purchase flow: Select product → Payment summary → Paystack checkout (web) → Return to app → PIN displayed + emailed
- Order history tab with receipt download option

Section 8: Wireframe Structure (Low-Fidelity)

8.1 Wireframe Conventions

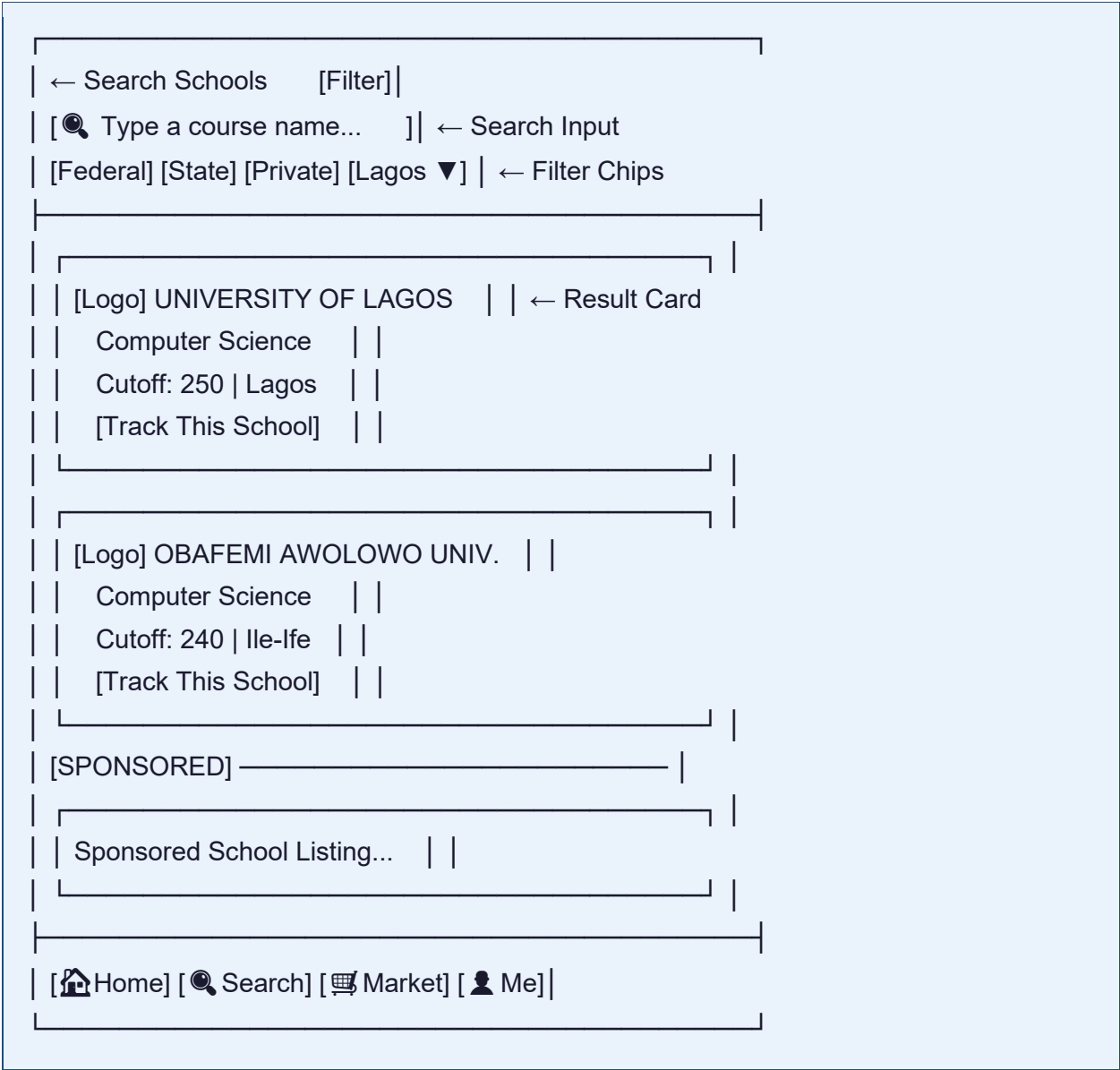
All wireframes below use ASCII-style structured layout descriptions suitable for handoff to designers. Each wireframe specifies element positioning, sizing hierarchy, and interaction states.

Mobile Screen Dimensions: 390 x 844px (iPhone 14 base), minimum 360px width support

8.2 Dashboard Wireframe

```
|
| [Enrollaa Logo] Hello, Chidi [🔔 3] | ← Top App Bar
| [Free User → UPGRADE pill]      |
|
|
|   |
|   | [🏫 UNILAG Computer Science] | | ← School Card 1
|   | Status: 🟡 MONITORING      | |
|   | Last checked: 3 mins ago   | |
|   |
|   |
|   | [🏫 OAU Computer Science]   | | ← School Card 2
|   | Status: 🟢 LIST PUBLISHED  | |
|   | [CHECK NOW →]              | |
|   |
|   |
|   | ————— AI PREDICTION ————— |
|   | [🟢 74%] High Chance at UNILAG CS | ← AI Widget
|   | [View Full Report →]        |
|   | ————— TODAY'S DEADLINES ————— |
|   | [FUTA Post-UTME · 2d] [ABU · 5d] → | ← Scroll
|   | ————— NATIONAL UPDATES ————— |
|   | • UNILAG releases 3rd batch list |
|   | • FUTMinna cutoff raised to 180  |
|   |
|
| [🏠 Home] [🔍 Search] [📊 Market] [👤 Me] | ← Bottom Nav
```

8.3 Search Screen Wireframe



Section 9: Technical Architecture

9.1 Architecture Overview

Enrollaa uses a modern, scalable microservice-ready monolith architecture in MVP phase, designed to decompose into microservices in Phase 2. The system is cloud-native (AWS/GCP/DigitalOcean) with containerized services.

9.2 Frontend Web Application (React)

- Framework: React 18 with TypeScript
- State Management: Zustand (lightweight, scales well) + React Query for server state
- Routing: React Router v6
- UI Library: Custom design system built on Tailwind CSS
- Payment UI: Paystack React SDK for payment modals
- Auth: JWT tokens stored in httpOnly cookies
- PWA: Service worker for offline capability and push notifications on web
- Build: Vite for fast development and production bundling
- Hosting: Vercel or Netlify with CDN

9.3 Mobile Application (React Native)

- Framework: React Native 0.73+ with Expo (managed workflow initially)
- Navigation: React Navigation v6 (Bottom Tabs + Stack Navigator)
- State: Zustand + React Query (shared logic with web where possible)
- Push Notifications: Expo Notifications → Firebase Cloud Messaging (Android) + APNs (iOS)
- In-app browser: expo-web-browser for Paystack payment redirect
- Local storage: AsyncStorage with encryption for sensitive profile data
- Offline support: React Query cache + AsyncStorage for dashboard data
- Analytics: Mixpanel or Amplitude for user behavior tracking

9.4 Backend Node.js + Express

- Runtime: Node.js 20 LTS
- Framework: Express.js with TypeScript
- Architecture: MVC + Service Layer + Repository Pattern
- Authentication: JWT (access token 15min + refresh token 30 days)
- Validation: Zod for schema validation on all endpoints

- File Storage: AWS S3 for uploaded documents and school logos
- Queue: Bull (Redis-backed) for background jobs (scraping, notifications, email)
- Caching: Redis for session cache, rate limiting, and frequently-accessed data
- Email: SendGrid or Mailgun for transactional emails
- SMS: Africa's Talking or Termii for SMS notifications
- Logging: Winston + Morgan with Datadog or Logtail

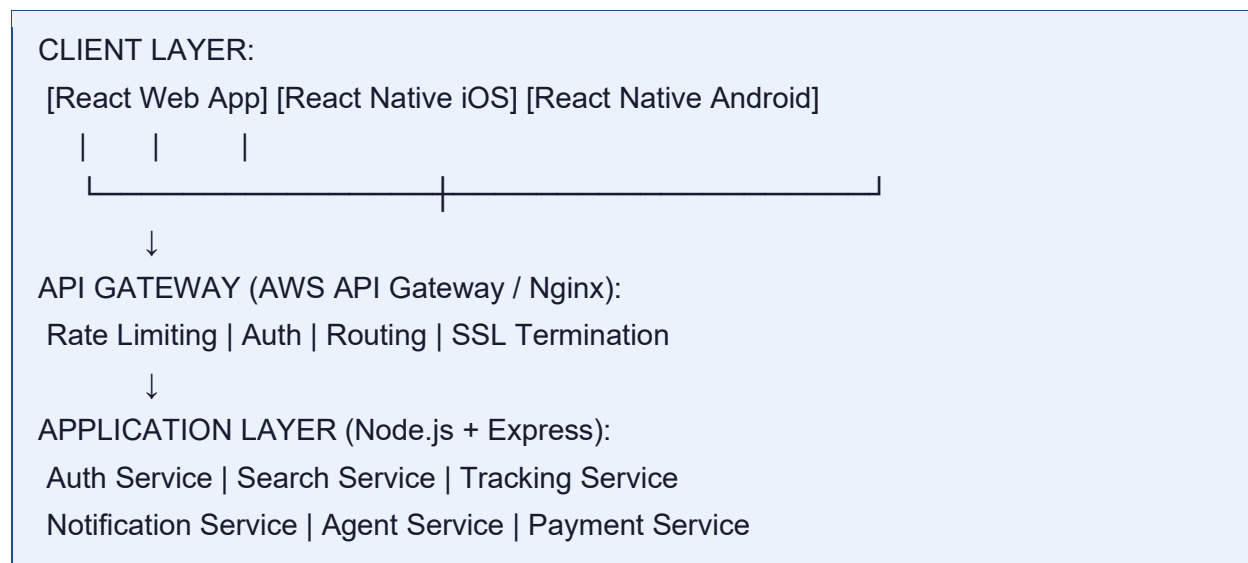
9.5 Database MongoDB

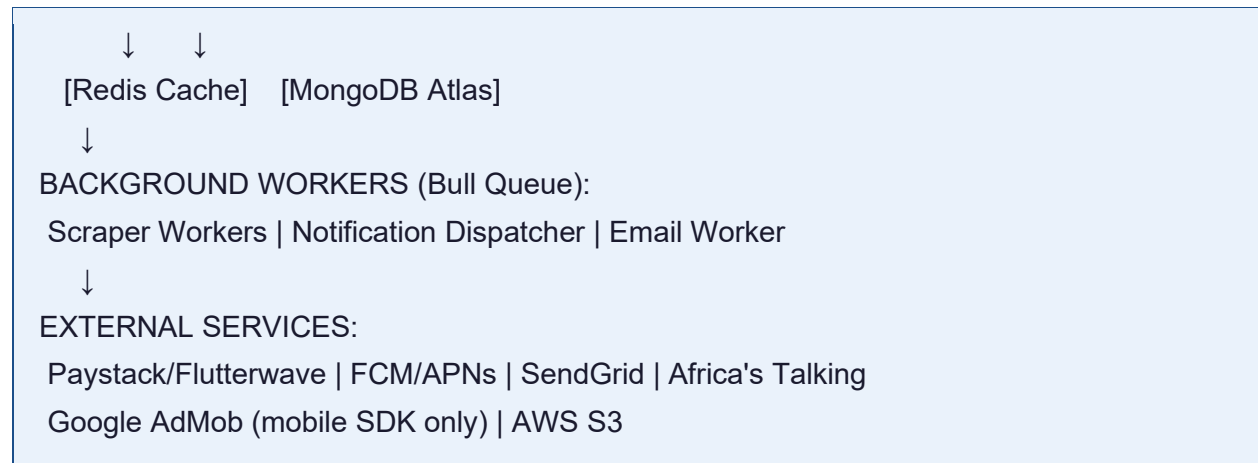
- Primary DB: MongoDB Atlas (M10 cluster for MVP, auto-scaling enabled)
- ODM: Mongoose with strict schema validation
- Indexes: Compound indexes on high-query fields (userId+schoolId, courseId+school)
- Backup: Daily automated backups to S3 with 30-day retention
- Cache layer: Redis sits in front of MongoDB for hot data

9.6 Scraping Infrastructure

- Engine: Puppeteer (headless Chrome) for JS-rendered sites; Cheerio for static HTML
- Orchestration: Bull queue with priority levels per school
- Scheduling: node-cron for timed scraping cycles
- Anti-block: Rotating residential proxies (Bright Data or Oxylabs) + randomized user agents
- Change Detection: SHA256 hash comparison of target DOM elements
- Failure handling: 3 retry attempts → fallback to manual review queue

9.7 Infrastructure Diagram





Section 10: Database Design

10.1 Users Collection

Collection: users

`_id`: ObjectId

`email`: String (unique, indexed, lowercase)

`phone`: String (unique, indexed)

`passwordHash`: String (bcrypt, 12 rounds)

`profile`: {

`firstName`: String

`lastName`: String

`stateOfOrigin`: String (ref: Nigerian states enum)

`stateOfPreference`: String

`jambScore`: Number (min: 0, max: 400)

`jambRegNumber`: String (unique, indexed)

`oLevelResults`: [{

`subject`: String

`grade`: String (enum: A1, B2, B3, C4, C5, C6, D7, E8, F9)

`examType`: String (enum: WAEC, NECO, NABTEB)

`year`: Number

 }]

`preferredCourse`: ObjectId (ref: courses)

`firstChoiceSchool`: ObjectId (ref: schools)

`secondChoiceSchool`: ObjectId (ref: schools)

}

`subscription`: {

`tier`: String (enum: free, premium)

`startDate`: Date

`endDate`: Date

`paystackSubscriptionCode`: String

}

`notificationPrefs`: { push: Boolean, email: Boolean, sms: Boolean }

`fcmTokens`: [String] (device push tokens)

isVerified: Boolean
isActive: Boolean
lastLoginAt: Date
createdAt: Date (auto)
updatedAt: Date (auto)

10.2 Schools Collection

Collection: schools

_id: ObjectId
name: String (indexed, text index for search)
abbreviation: String (e.g., 'UNILAG', 'OAU')
jambInstitutionCode: String (unique)
type: String (enum: Federal, State, Private, Polytechnic, College of Education)
state: String
city: String
address: String
website: String (official URL used by scraper)
admissionPortalUrl: String (specific admission page URL)
logoUrl: String (AWS S3 URL)
accreditationStatus: String (enum: Accredited, Provisional, None)
scrapingConfig: {
 isScrapingEnabled: Boolean
 targetSelectors: Object (CSS selectors for admission data)
 lastSuccessfulScrape: Date
 scrapeFrequencyMinutes: Number
 requiresAuth: Boolean
 usesPuppeteer: Boolean
}
contacts: { phone: String, email: String }
createdAt: Date
updatedAt: Date

10.3 Courses Collection

Collection: courses

`_id`: ObjectId

`name`: String (text indexed, e.g., 'Computer Science')

`aliases`: [String] (e.g., ['Computing', 'Comp Sci'])

`faculty`: String (e.g., 'Science', 'Engineering')

`jambSubjectCombination`: [String] (required JAMB subjects)

`oLevelRequirements`: [{

`subject`: String

`minimumGrade`: String

`isCompulsory`: Boolean

}]

`nuc_approved`: Boolean

`createdAt`: Date

// Indexes: text index on name + aliases; compound on faculty

10.4 Admissions Collection

Collection: admissions

`_id`: ObjectId

`schoolId`: ObjectId (ref: schools, indexed)

`courseId`: ObjectId (ref: courses, indexed)

`academicYear`: String (e.g., '2024/2025')

`cutoffMark`: Number

`postUtmeDate`: Date

`postUtmeRegistrationDeadline`: Date

`admissionListStatus`: String (enum: not_published, batch1, batch2, batch3, supplementary, closed)

`admissionListUrl`: String (direct link to published list)

`admittedCount`: Number

`appliedCount`: Number

screeningRequirements: String (free text / HTML)
notes: String
isVerified: Boolean (manually verified by agent)
dataSource: String (enum: scraped, manual, official_api)
lastUpdatedAt: Date (indexed)
createdAt: Date

10.5 Notifications Collection

Collection: notifications

_id: ObjectId
userId: ObjectId (ref: users, indexed)
type: String (enum: admission_update, deadline_reminder, status_change, system, agent_message)
title: String
body: String
relatedSchoolId: ObjectId (optional)
relatedAdmissionId: ObjectId (optional)
isRead: Boolean (default: false, indexed)
deliveryChannels: { push: Boolean, email: Boolean, sms: Boolean }
deliveredAt: { push: Date, email: Date, sms: Date }
createdAt: Date (indexed)

10.6 Agents Collection (Internal Only)

Collection: agents

_id: ObjectId
userId: ObjectId (ref: users agents also have user accounts with elevated role)
employeeId: String (internal HR reference)
name: String
email: String (company email, @Enrollaa.ng)

```
role: String (enum: agent, senior_agent, admin)
isActive: Boolean
activeConversations: Number (current open tickets)
maxConcurrentChats: Number (default: 10)
totalConversationsHandled: Number
averageResponseTimeMinutes: Number
shift: { start: String ('08:00'), end: String ('17:00'), timezone: 'Africa/Lagos' }
hiredAt: Date
createdAt: Date
```

10.7 Transactions Collection

Collection: transactions

```
_id: ObjectId
userId: ObjectId (ref: users, indexed)
type: String (enum: subscription, marketplace_purchase)
status: String (enum: pending, success, failed, refunded, indexed)
amount: Number (in kobo for Paystack, e.g., 400000 = ₦4,000)
currency: String (default: NGN)
paystackReference: String (unique, indexed)
paystackTransactionId: String
metadata: Object {
  subscriptionTier: String,
  marketplaceProductId: ObjectId,
  productName: String
}
paidAt: Date
createdAt: Date
```

10.8 Marketplace Products Collection

Collection: marketplace_products


```
_id: ObjectId
name: String (e.g., 'JAMB ePIN', 'WAEC Result PIN')
slug: String (unique, e.g., 'jamb-epin')
description: String
price: Number (in kobo)
vendor: {
  name: String (e.g., 'JAMB', 'WAEC'),
  isVerified: Boolean,
  apiEndpoint: String (vendor API for PIN generation)
}
category: String (enum: jamb, waec, neco, other)
isAvailable: Boolean
stockCount: Number (null = unlimited)
imageUrl: String
deliveryMethod: String (enum: api_generated, manual_fulfillment)
deliveryTimeSeconds: Number (expected delivery time)
createdAt: Date
```

Section 11: Monitoring System (Critical)

11.1 Overview

The monitoring system is the technical heart of Enrollaa. It continuously tracks official Nigerian university websites and JAMB portals, detects changes, and propagates updates to users in real-time. This is what differentiates Enrollaa from every other platform in the market.

11.2 Scraping Architecture

Tool Stack

- Puppeteer (headless Chromium): For JavaScript-heavy portals (most Nigerian university sites use jQuery-based systems)
- Cheerio: For lighter static HTML pages where Puppeteer is overkill
- Playwright (future): For sites with advanced bot detection
- Proxy rotation: Bright Data residential proxies to avoid IP blocks

Change Detection Algorithm

1. Scraper fetches target URL and extracts specific DOM elements (defined per school in scrapingConfig)
2. Content of target elements is hashed using SHA256
3. New hash is compared to the stored hash in Redis (fast comparison)
4. If hash differs: trigger update pipeline
5. If hash same: log 'no change', update lastChecked timestamp
6. If fetch fails: increment failureCount; at 3 failures, move to manual review queue

11.3 Cron Schedule Design

Condition	Schedule
Peak season (Aug–Oct)	Every 15 minutes for Top 50 schools
Peak season (Aug–Oct)	Every 30 minutes for schools 51–150
Off-peak season	Every 4 hours for all schools
JAMB.gov.ng portal	Every 10 minutes year-round
Supplementary list period	Every 5 minutes for schools with list status = pending
Manual review queue	Agents check flagged schools every 2 hours

11.4 Scraper Worker Code Pattern

```
// ScraperWorker.ts (simplified)
async function scrapeSchool(schoolId: string) {
  const school = await SchoolRepo.findById(schoolId);
  const { admissionPortalUrl, scrapingConfig } = school;

  let content: string;
  if (scrapingConfig.usesPuppeteer) {
    const browser = await puppeteer.launch({ headless: true });
    const page = await browser.newPage();
    await page.setUserAgent(randomUserAgent());
    await page.goto(admissionPortalUrl, { waitUntil: 'networkidle2' });
    content = await page.$eval(scrapingConfig.targetSelectors.main, el => el.innerHTML);
    await browser.close();
  } else {
    const html = await axios.get(admissionPortalUrl);
    const $ = cheerio.load(html.data);
    content = $(scrapingConfig.targetSelectors.main).html();
  }

  const newHash = sha256(content);
  const cachedHash = await redis.get(`hash:school:${schoolId}`);

  if (newHash !== cachedHash) {
    await redis.set(`hash:school:${schoolId}`, newHash);
    await AdmissionUpdateQueue.add({ schoolId, content, detectedAt: new Date() });
  }
}
```

11.5 Update Pipeline

7. Change detected → raw content sent to AdmissionUpdateQueue
8. Parser worker processes content: extracts cutoff marks, list status, dates using regex + NLP
9. Parsed data validated against Admission schema
10. Admission document in MongoDB updated atomically

11. Affected users identified (those tracking this school)
12. Notification queue populated for affected users
13. Push/email/SMS dispatched per user notification preferences

Section 12: AI System Design

12.1 AI Architecture Philosophy

Enrollaa's AI system is designed to be practical, explainable, and trainable on Nigerian-specific data. We avoid black-box models in favor of transparent, interpretable logic that builds user trust. The system starts with rule-based ML and evolves into neural models as data grows.

12.2 Admission Prediction Engine

Input Features (per prediction request)

- Student JAMB score
- Student O'Level grades (individual subjects)
- State of origin vs. school state (catchment area bonus applies at many schools)
- Target course competitiveness index (derived from historical admitted:applied ratio)
- Historical cutoff trends (3-year average, year-over-year delta)
- Number of applicants currently tracked for this school-course (proxy for competition level)

Model V1 Logistic Regression / Decision Tree (MVP)

For MVP, we use a weighted scoring model with logistic regression for probability output:

$$\text{AdmissionProbability} = \text{sigmoid}(w1*\text{jambScore} + w2*\text{oLevelScore} + w3*\text{catchmentBonus} + w4*\text{historicalAcceptanceRate} + w5*\text{trendFactor} + \text{bias})$$

Where:

`oLevelScore` = weighted sum of relevant subject grades

`catchmentBonus` = +0.05 if student state matches school catchment state

`historicalAcceptanceRate` = (avg admitted / avg applied) over 3 years

`trendFactor` = +/- based on whether cutoff is rising or falling

Model V2 Random Forest (Phase 2)

As data accumulates (target: 10,000+ historical admission records), upgrade to a Random Forest ensemble that captures non-linear relationships between features, including course-specific and school-specific patterns.

Model V3 Fine-tuned LLM for Recommendations (Phase 3)

Use a fine-tuned language model to generate natural language explanations of predictions and personalized action plans (e.g., 'Your JAMB score is 5 points below UNILAG's typical cutoff. Based on current trends, consider OAU or FUTA as safer alternatives for Computer Science').

12.3 Recommendation Engine

When a student's probability at their preferred school is below 60%, the system automatically generates alternative recommendations:

- Same course, lower-cutoff schools in the same or adjacent states
- Related courses at preferred school with higher admission probability
- Ranked list of 5 alternatives with probability scores for each
- Historical data on whether alternatives are 'rising' or 'declining' in admissions

12.4 NLP Pipeline for Scraped Content

Raw scraped HTML from school websites must be parsed into structured data:

- 14. HTML cleaning and text extraction
- 15. Named Entity Recognition (NER) to identify school names, course names, cutoff marks
- 16. Pattern matching for Nigerian admission terminology (batch list, supplementary, post-UTME)
- 17. Date extraction and normalization to ISO format
- 18. Confidence score per extracted field (low confidence → manual review flag)

12.5 How AI Improves Over Time

Phase	AI Evolution
Year 1	Logistic regression on 3 years of historical JAMB data; manual weight tuning
Year 2	Random Forest trained on platform-generated admission outcomes; automated retraining monthly
Year 3	Gradient boosting (XGBoost) + user feedback loop (admitted students confirm outcomes)
Year 4+	Neural network with course-school embeddings; real-time A/B testing of prediction models

Section 13: Internal Agent Support System

13.1 System Design Principles

The agent system is built on a zero-trust-to-external model. Every rule is designed to eliminate the possibility of fraud originating from within the support channel.

CARDINAL RULES ENFORCED AT SYSTEM LEVEL:

1. No agent account can initiate a payment request to a user
2. No agent can share a personal phone number, email, or external link
3. No user can pay any agent outside the platform payment flows only through Paystack
4. All conversations are stored permanently and auditable
5. Admin can terminate any agent session in real-time
6. Agents cannot see or access full payment details (PCI-DSS compliant view)

13.2 Agent Roles & Permissions (MVP)

Role	Permissions
Agent (x2 in MVP)	Respond to user chats, view user profile, update conversation status, escalate to admin
Admin (x1)	View all conversations, monitor agents in real-time, terminate sessions, manage agent accounts, view all transactions
System Bot	Handle automated responses, ticket routing, off-hours messages

13.3 Agent Dashboard Features (Web Interface)

- Inbox with three columns: Open | In Progress | Resolved
- Each ticket shows: User name, JAMB score, issue type, time since last message, SLA timer
- Conversation panel with full chat history
- User sidebar: Profile details, tracked schools, subscription status, transaction history
- Quick-reply templates for common queries (cutoff mark questions, ePIN delivery, etc.)
- Escalation button: Forward to admin with notes
- 'Send Official Resource' button: Agents can only share pre-approved links from the platform's curated resource library

13.4 User-Facing Support Experience

- Access via 'Support' tab in bottom navigation
- New conversation requires topic selection (reduces misdirected queries)
- Automated bot handles: FAQs, status checks, ePIN delivery reissue (reduces agent load by ~40%)
- Human agent connects within 2 hours during business hours (08:00–18:00 WAT)
- Off-hours: Automated response with expected reply time; urgent issues flagged for next-day priority
- User satisfaction rating (1–5 stars) after conversation closure

13.5 Conversation Logging & Compliance

- All messages stored in the conversations collection in MongoDB
- Retention policy: 2 years minimum
- Admin can search conversations by userId, agentId, date range, keywords
- Automated scanning for prohibited content (phone numbers, bank account numbers, external payment links)
- Violation alerts: System flags and notifies admin if an agent sends content matching prohibited patterns

Section 14: Marketplace System

14.1 Marketplace Overview

The marketplace allows students to purchase official educational service PINs directly within the Enrollaa platform, eliminating the need to visit untrustworthy third-party vendors. All purchases are processed through Paystack/Flutterwave and digitally delivered within the app.

14.2 JAMB ePIN Purchase Flow

19. User taps 'Buy JAMB ePIN' from Dashboard or Marketplace tab
20. Product details screen: Price (e.g., ₦3,500), what ePIN is used for, step-by-step guide
21. User taps 'Buy Now'
22. On mobile: expo-web-browser opens Paystack payment page (iOS App Store compliant payment on web, not in-app)
23. On web: Paystack modal opens inline
24. User completes payment (card, bank transfer, USSD)
25. Paystack sends webhook to Enrollaa backend (POST /webhooks/paystack)
26. Backend verifies webhook signature, confirms transaction
27. Backend calls JAMB ePIN vendor API (or internal stock system) to generate/retrieve PIN
28. Transaction record created in MongoDB with PIN stored encrypted
29. User sees PIN displayed in-app immediately
30. PIN also sent to user's registered email
31. Receipt available in 'My Orders' section

14.3 Payment Security

- Paystack webhook verification: HMAC-SHA512 signature validation on every webhook
- PINs stored encrypted at rest (AES-256) in MongoDB
- PINs only decrypted and displayed to the purchasing user
- All transactions logged with Paystack reference ID for reconciliation
- Refund policy: Automated refund via Paystack API for failed PIN delivery within 24 hours

14.4 Vendor Verification Framework

Product	Verification Method
JAMB ePIN	Official JAMB API partnership (or licensed bulk PIN distributor)
WAEC Result PIN	Official WAEC vendor agreement; bulk PIN purchase + inventory management

NECO PIN	NECO licensed vendor integration
Other services	Whitelist-only; must pass vendor verification checklist before listing

Section 15: Monetization Strategy

15.1 Revenue Streams

Revenue Stream	Description
Subscription (Premium)	₦4,000–₦5,000 /year (~\$2.50–\$3.20 USD). Primary revenue driver.
Marketplace Margin	5–15% margin on JAMB ePIN and WAEC PIN sales. Volume-driven.
Google AdMob (Free Tier)	Banner and interstitial ads for non-paying users only. Minimal, non-intrusive.
B2B School Partnerships (Phase 2)	Schools pay for verified 'Official School Page' status and direct student outreach.
Data Insights (Phase 3)	Anonymized, aggregated admission trend reports sold to universities and education policymakers.

15.2 Free vs Premium Tier Breakdown

Feature	Free Tier	Premium Tier
Course search (all schools)	YES Free	YES Premium
Basic school details	YES Free	YES Premium
Dashboard (2 schools)	YES Free (2 schools max)	YES Unlimited schools
Real-time push notifications	NO Free (delayed, daily digest)	YES Premium (instant)
AI admission prediction	NO Free (score hidden, teaser shown)	YES Full score + report
Agent priority support	NO Free (standard queue, 24hr SLA)	YES 2hr SLA
Marketplace access	NO Free	YES Premium
Ads	YES Free (AdMob)	NO Premium (ad-free)
Export/PDF report	NO Free	YES Premium

15.3 Subscription Pricing Rationale

At ~~₦4,000–₦5,000~~ per year, the subscription is priced at approximately the cost of one fraudulent agent interaction making the value proposition immediate and tangible to Nigerian families. At target of 10,000 premium users, annual subscription revenue = ~~₦40–₦50~~ million (~\$25,000–\$31,000 USD). This funds operations and team salaries in Phase 1.

15.4 Google AdMob Integration

- SDK: Google AdMob React Native SDK
- Ad formats: Banner ads (bottom of screen, non-intrusive) + interstitial on search results (max 1 per session)
- Ad gating: AdMob SDK is initialized only when `user.subscription.tier === 'free'`
- Premium users: AdMob SDK is not initialized at all no tracking, no ads
- COPPA compliance: Age gate enforced; no personalized ads for users under 18

Section 16: User Flow Diagrams

16.1 New User Acquisition Flow

LANDING PAGE → [Start Free] → PHONE/EMAIL REGISTRATION → OTP VERIFICATION
↓
PROFILE SETUP (JAMB Score + Course + School + O'Level)
↓
NOTIFICATION PERMISSION REQUEST
↓
DASHBOARD (Free Tier)
↓
[AI Prediction teaser shown] → [Upgrade prompt]
↓
PAYSTACK PAYMENT (Web / Web-Browser) → SUBSCRIPTION ACTIVATED
↓
PREMIUM DASHBOARD (Full features unlocked)

16.2 Returning User Admission Alert Flow

CRON JOB triggers school scrape (every 15–30 mins)
↓
CHANGE DETECTED at school portal
↓
Admission record UPDATED in MongoDB
↓
AFFECTED USERS identified (those tracking this school)
↓
NOTIFICATION dispatched: Push + Email + SMS (per prefs)
↓
USER opens notification → SCHOOL DETAIL SCREEN
↓
STATUS UPDATE shown: 'Batch 2 list published Check your name'
↓

[Check Admission List] → External school portal link (verified)

16.3 Marketplace Purchase Flow

MARKETPLACE TAB → Select Product (e.g., JAMB ePIN)
↓
PRODUCT DETAIL SCREEN → [Buy Now] tapped
↓
PAYSTACK CHECKOUT opens (web browser / modal)
↓
PAYMENT COMPLETED → Paystack Webhook → Backend verified
↓
VENDOR API called → PIN retrieved
↓
PIN displayed IN-APP (30–60 seconds post-payment)
↓
PIN emailed to registered email
↓
Receipt available in 'My Orders'

Section 17: API Design

17.1 API Conventions

- Base URL: <https://api.Enrollaa.ng/v1>
- Authentication: Bearer token in Authorization header
- Response format: JSON with { success: boolean, data: object, message: string, errors: array }
- Rate limiting: 100 requests/minute for authenticated users; 20/minute for unauthenticated
- All dates: ISO 8601 (UTC)

17.2 Authentication Endpoints

Endpoint	Description
POST /auth/register	Register new user with email/phone + password. Returns OTP trigger.
POST /auth/verify-otp	Verify OTP. Returns { accessToken, refreshToken }.
POST /auth/login	Login with email/phone + password. Returns tokens.
POST /auth/refresh	Refresh access token using refresh token.
POST /auth/logout	Invalidate refresh token.
POST /auth/forgot-password	Send password reset link to email.
PATCH /auth/reset-password	Reset password with token from email.

17.3 User Profile Endpoints

Endpoint	Description
GET /users/me	Get current user's full profile.
PATCH /users/me	Update profile fields (JAMB score, O'Level, preferences).
GET /users/me/tracked-schools	Get all schools the user is tracking.
POST /users/me/tracked-schools	Add a school to tracking (premium: unlimited; free: max 2).
DELETE /users/me/tracked-schools/:schoolId	Remove a school from tracking.

17.4 School & Admission Endpoints

Endpoint	Description
GET /schools	Search schools. Query params: course, state, type, sort.

GET /schools/:id	Get school detail with all courses and cutoff data.
GET /schools/:id/admissions	Get admission records for a school. Query: year.
GET /courses	Search courses. Query: q (name search).
GET /courses/:id	Get course details with O'Level requirements.

17.5 AI Prediction Endpoints

Endpoint	Description
GET /predictions/my-chances	Returns admission probability for user's tracked schools. Auth required.
GET /predictions/compare?schools=id1,id2	Compare admission chances across multiple schools.
GET /predictions/recommendations	Get 5 alternative school-course recommendations.
POST /predictions/custom	Run prediction for arbitrary JAMB score + course + school.

17.6 Notification Endpoints

Endpoint	Description
GET /notifications	Get user notifications. Query: type, isRead, page, limit.
PATCH /notifications/:id/read	Mark notification as read.
PATCH /notifications/read-all	Mark all as read.
DELETE /notifications/:id	Delete a notification.
POST /notifications/register-token	Register FCM push token for device.

17.7 Payment & Subscription Endpoints

Endpoint	Description
POST /payments/initialize-subscription	Create Paystack subscription checkout. Returns payment URL.
GET /payments/verify/:reference	Verify a Paystack transaction by reference.
POST /webhooks/paystack	Receive Paystack webhook events (public endpoint, HMAC verified).
GET /payments/transactions	Get user transaction history.
POST /payments/marketplace/purchase	Initiate marketplace product purchase.

17.8 Agent Support Endpoints

Endpoint	Description
----------	-------------

GET /support/conversations	Get user's support conversations.
POST /support/conversations	Create new support conversation with topic.
GET /support/conversations/:id/messages	Get messages in a conversation.
POST /support/conversations/:id/messages	Send a message.
PATCH /support/conversations/:id/close	Close a conversation (user or agent).
POST /support/conversations/:id/rate	Rate conversation (1–5 stars).
GET /agent/inbox	Agent: Get all conversations in agent's queue. [Agent role required]
PATCH /agent/conversations/:id/assign	Agent: Assign conversation to self. [Agent role]

Section 18: Codebase Folder Structure

18.1 Backend Node.js + Express + TypeScript

```
Enrollaa-backend/
├── src/
│   ├── config/      # App config, env vars, DB connection
│   │   ├── database.ts
│   │   ├── redis.ts
│   │   └── env.ts
│   ├── api/
│   │   ├── routes/  # Express routers per domain
│   │   │   ├── auth.routes.ts
│   │   │   ├── users.routes.ts
│   │   │   ├── schools.routes.ts
│   │   │   ├── admissions.routes.ts
│   │   │   ├── predictions.routes.ts
│   │   │   ├── notifications.routes.ts
│   │   │   ├── payments.routes.ts
│   │   │   ├── marketplace.routes.ts
│   │   │   └── support.routes.ts
│   │   ├── controllers/ # Route handlers (thin layer)
│   │   ├── middleware/  # Auth, rate limit, validation, error handler
│   │   └── validators/  # Zod schemas per endpoint
│   ├── services/      # Core business logic
│   │   ├── auth.service.ts
│   │   ├── user.service.ts
│   │   ├── school.service.ts
│   │   ├── admission.service.ts
│   │   ├── prediction.service.ts
│   │   ├── notification.service.ts
│   │   ├── payment.service.ts
│   │   ├── marketplace.service.ts
│   │   └── support.service.ts
│   └── repositories/  # MongoDB queries (Mongoose)
```

```
| |— models/      # Mongoose schemas
| |— workers/     # Bull queue workers
| |   |— scraper.worker.ts
| |   |— notification.worker.ts
| |   |— email.worker.ts
| |   |— sms.worker.ts
| |— scrapers/    # School-specific scraper configs
| |   |— base.scraper.ts
| |   |— jamb.scraper.ts
| |   |— schools/  # Per-school scraping configs
| |— ai/         # ML prediction logic
| |   |— prediction.model.ts
| |   |— recommendation.engine.ts
| |— utils/      # Shared utilities
| |— app.ts      # Express app entry point
|— tests/       # Jest unit + integration tests
|— scripts/     # DB seed, migration scripts
|— .env.example
|— Dockerfile
|— docker-compose.yml
|— package.json
```

18.2 Frontend Web React + TypeScript

```
Enrollaa-web/
|— src/
| |— pages/
| |   |— Landing/
| |   |— Auth/    # Login, Register, OTP
| |   |— Onboarding/
| |   |— Dashboard/
| |   |— Search/
| |   |— SchoolDetail/
| |   |— Predictions/
| |   |— Notifications/
```

```
| | └─ Support/
| | └─ Marketplace/
| | └─ Account/
| └─ components/    # Reusable UI components
| | └─ ui/          # Base: Button, Input, Card, Modal
| | └─ layout/      # Navbar, Footer, Sidebar
| | └─ domain/      # SchoolCard, AdmissionBadge, etc.
| └─ hooks/         # Custom React hooks
| └─ stores/        # Zustand stores
| └─ services/      # API client functions
| └─ types/         # TypeScript type definitions
| └─ utils/
└─ public/
└─ vite.config.ts
```

18.3 Mobile App React Native + Expo

```
Enrollaa-mobile/
└─ src/
| └─ screens/      # One folder per screen
| | └─ Auth/
| | └─ Onboarding/
| | └─ Dashboard/
| | └─ Search/
| | └─ SchoolDetail/
| | └─ Predictions/
| | └─ Notifications/
| | └─ Support/
| | └─ Marketplace/
| | └─ Account/
| └─ navigation/   # React Navigation setup
| └─ components/   # Shared mobile components
| └─ hooks/
| └─ stores/       # Zustand (shared logic with web)
| └─ services/     # API client
```

```
| └─ utils/  
├─ app.json    # Expo config  
└─ eas.json    # EAS Build config
```

Section 19: Development Roadmap

19.1 Phase 1 MVP (Months 1–4)

Goal: Validated, working product with core admission tracking, basic AI, 2 agents, marketplace, and subscription billing.

Month	Deliverable
Month 1	Backend setup (Node/Express/MongoDB), user auth, school data seeding (100 schools), basic search API
Month 2	Scraper system (20 top schools), change detection, notification pipeline, mobile app onboarding screens
Month 3	Dashboard, AI prediction V1 (logistic regression), agent chat system, Paystack integration
Month 4	Marketplace (JAMB ePIN + WAEC PIN), AdMob integration, beta testing (100 users), bug fixes, App Store submission

19.2 Phase 2 Growth (Months 5–10)

- Expand scraping to 200 schools
- AI Model V2 (Random Forest) with improved prediction accuracy
- School Partnership Program (official school pages)
- Parent account: Link parent to child's profile
- Advanced analytics dashboard for premium users
- B2B: Sell admission data packages to schools
- Referral program to accelerate user acquisition
- Android and iOS App Store optimization (ASO)

19.3 Phase 3 Scale (Months 11–18)

- West African expansion: Ghana (WAEC/WASSCE focused), then Kenya
- Scholarship discovery engine integrated into recommendations
- Advanced AI: Fine-tuned LLM for natural language admission counseling
- API product: Allow third-party education apps to query Enrollaa data
- Enterprise dashboard: Ministry of Education/JAMB partnership

Section 20: Risk Analysis

20.1 Risk Register

Risk	Severity	Mitigation
Scraping blocked by schools	HIGH	Use proxy rotation + Playwright; maintain manual fallback; build school relationships for official data access
Data accuracy errors	HIGH	Agent verification layer for critical data; confidence scores on all scraped data; user feedback loop to flag errors
Paystack payment failures	MEDIUM	Implement retry logic; provide alternative payment methods (bank transfer); customer support escalation path
User data breach	HIGH	Encrypt sensitive data at rest and in transit; security audits quarterly; no storage of full card data (Paystack handles PCI)
Competitor copies model	MEDIUM	Build data moat (3+ years of historical data by Year 2); trust moat through verified ecosystem; agent quality
Low subscription conversion	MEDIUM	A/B test paywall placement; offer student discount; annual vs monthly option; free trial for premium
Agent misconduct	LOW (mitigated)	Real-time monitoring of all chats; automated pattern detection; zero direct payment channels for agents
App Store rejection (iOS)	MEDIUM	Payments done via web browser, not in-app purchases App Store compliant; reviewed by legal before submission

Section 21: Scalability Plan

21.1 Infrastructure Scaling

- Horizontal scaling: Node.js app behind AWS Application Load Balancer; auto-scaling group scales from 2 to 20 instances based on CPU/memory
- Database scaling: MongoDB Atlas M10 → M30 → M60 tiers; read replicas for heavy read queries (search, prediction)
- Redis Cluster: Move from single Redis instance to Redis Cluster when data exceeds 8GB
- CDN: CloudFront for static assets; reduces latency for Nigerian users from Lagos, Abuja, Port Harcourt simultaneously
- Scraping fleet: Scale from 2 to 20 scraper worker containers during peak admission season (Aug–Oct) using Docker + ECS

21.2 Database Scalability

- Sharding strategy: Shard users collection by stateOfOrigin for geographic distribution
- TTL indexes: Notifications auto-expire after 90 days to manage collection size
- Archival: Admission records older than 5 years archived to S3 cold storage
- Index optimization: Explain plan analysis monthly; compound indexes reviewed quarterly

21.3 AI/ML Scaling

- Model serving: Move from in-process model to dedicated ML microservice (FastAPI + Python) by Phase 2
- Batch predictions: Pre-compute predictions for all user-school combinations nightly during off-peak hours
- Model retraining: Automated retraining pipeline with new admission cycle data each year

21.4 Multi-Country Scaling

- Multi-tenancy: Data separated by country (Nigeria, Ghana, Kenya) at the database level using tenantId field
- Localization: i18n support built from Day 1 (English, Pidgin, Hausa, Yoruba, Igbo in Phase 3)
- Payment localization: Flutterwave supports 30+ African currencies; extend marketplace per country

Section 22: Go-To-Market Strategy

22.1 Launch Strategy Nigeria First

Enrollaa launches in Nigeria focused on JAMB season (February–November). The highest concentration of prospective users is active between March and September, with the admission list tracking peak in June–October.

22.2 Pre-Launch (Month 1–3)

- Build waitlist via landing page: 'Get Early Access' form with email capture
- Content marketing: Publish weekly blog posts on 'How JAMB cutoff marks work', 'Top Engineering schools in Nigeria by cutoff', etc. target SEO keywords
- WhatsApp community: Build a monitored WhatsApp community for beta testers (NOT for admission help for product feedback only)
- University campus ambassadors: Recruit 10 student ambassadors at UNILAG, OAU, UNIBEN, ABU, UNIZIK each earns ₦500 per premium referral

22.3 Launch (Month 4)

- App Store + Google Play Store launch with App Store Optimization (ASO)
- Press release to Nigerian tech media: TechCabal, Techpoint, Nairametrics
- Twitter/X campaign: '#Enrollaa Know Your Admission Status in Real Time'
- YouTube explainer video: 2-minute demo of the platform (produced on budget using Nigerian student creators)
- Free tier with aggressive onboarding: Get 1,000 users in Week 1 through campus ambassador network

22.4 Growth (Month 5+)

- Referral program: Each user gets a unique code. If friend subscribes, both get 1 month free premium.
- Partnership with JAMB CBT centers: Distribute pamphlets and offer Enrollaa demo during registration period
- Parent targeting: Facebook Ads targeting parents in Nigeria aged 35–55 with teenagers
- SMS campaign: Bulk SMS to JAMB candidates (via licensed data provider) with download link during registration season
- NECO/WAEC result season: Push 'Check Your Result + Get Admission Intel' campaign

22.5 KPIs for Launch

Milestone	Target
Month 1 (Beta)	100 active users, 5% premium conversion
Month 4 (Public Launch)	1,000 registered users, 8% premium conversion
Month 6	5,000 users, 10% premium = 500 paid subscriptions = ₦2M ARR
Month 12	20,000 users, 12% premium = 2,400 paid = ₦9.6–12M ARR

Section 23: Future Features

23.1 International Expansion

- Ghana (Phase 3): Track WAEC BECE/WASSCE results + Ghana Tertiary admissions (KNUST, UG, UCC)
- Kenya (Phase 3): KCSE results + KUCCPS admission tracking
- South Africa (Phase 4): NSC results + NSFAS funding tracking for HEI applications
- Pan-African expansion goal: 10 countries by Year 5, powered by country-specific scraper configs

23.2 Scholarship Discovery Engine

- Database of active Nigerian and international scholarships (Federal Government, state, corporate, international)
- Matching engine: 'Based on your profile, you qualify for these 7 scholarships 2 deadlines this month'
- Application tracking for scholarship applications
- Integration with scholarship portals (Scholarships.gov.ng, NNPC scholarship, etc.)

23.3 Advanced AI Features

- Admission Counselor Chatbot: Fine-tuned LLM that can answer 'What subjects do I need for Medicine at UNILAG?' in conversational language
- Essay Helper: AI-powered personal statement and application essay assistance for international university applications
- Score Improvement Predictor: 'If you improve your JAMB score by 20 points next year, your admission chances at UNILAG CS rise from 74% to 91%'
- Peer Comparison (anonymized): 'Students with similar profiles as yours 67% were admitted in this cycle'

23.4 Post-Admission Features

- Acceptance letter upload and verification
- School fee payment tracker
- Hostel accommodation marketplace (verified school hostels only)
- Textbook marketplace (buy/sell used textbooks within university communities)
- Alumni network: Connect with students at your school for peer support

23.5 B2B Product Line

- University Admin Dashboard: Schools pay for a portal to publish official admission info directly to Enrollaa (eliminates scraping need for participating schools)
- JAMB Partnership API: Official data feed from JAMB for real-time CAPS status (in discussions)
- Ministry of Education Data Reports: Aggregated, anonymized admission trend reports for policy decisions

DOCUMENT SUMMARY

This document covers all 23 sections of the Enrollaa product specification, from product vision through technical architecture, AI design, agent system, marketplace, monetization, API design, codebase structure, roadmap, risk analysis, go-to-market strategy, and future features.

Version: 1.0 | Status: MVP Specification | Audience: Founders, Developers, Designers, Investors

Next Steps: Proceed to technical sprint planning and designer handoff.